

Trigger Points

Sara Modlin-Tucker D.O.

Objectives:

Describe the history of trigger point theory development.
Define and understand myofascial trigger points.

Trigger Point Definition

Glossary of Osteopathic Terminology

A small hypersensitive site that, when stimulated, consistently produces a reflex mechanism that gives rise to referred pain and/or other manifestations in a consistent reference zone that is consistent from person to person.

Travell & Simons (1983):

“A hyperirritable spot in skeletal muscle associated with a hypersensitive palpable nodule in a taut band. The spot is painful on compression and can give rise to characteristic referred pain, motor dysfunction, and autonomic phenomena.”

Tender Points?

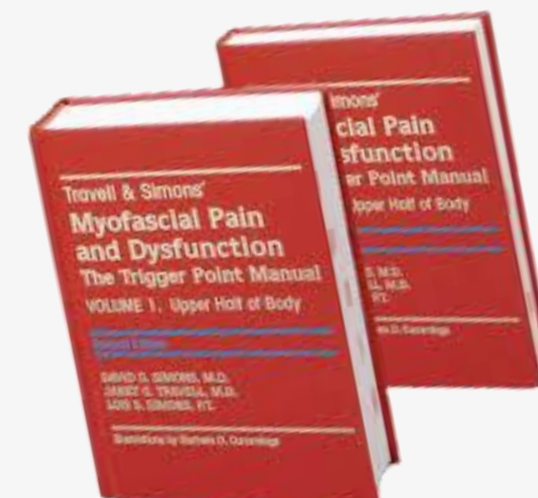
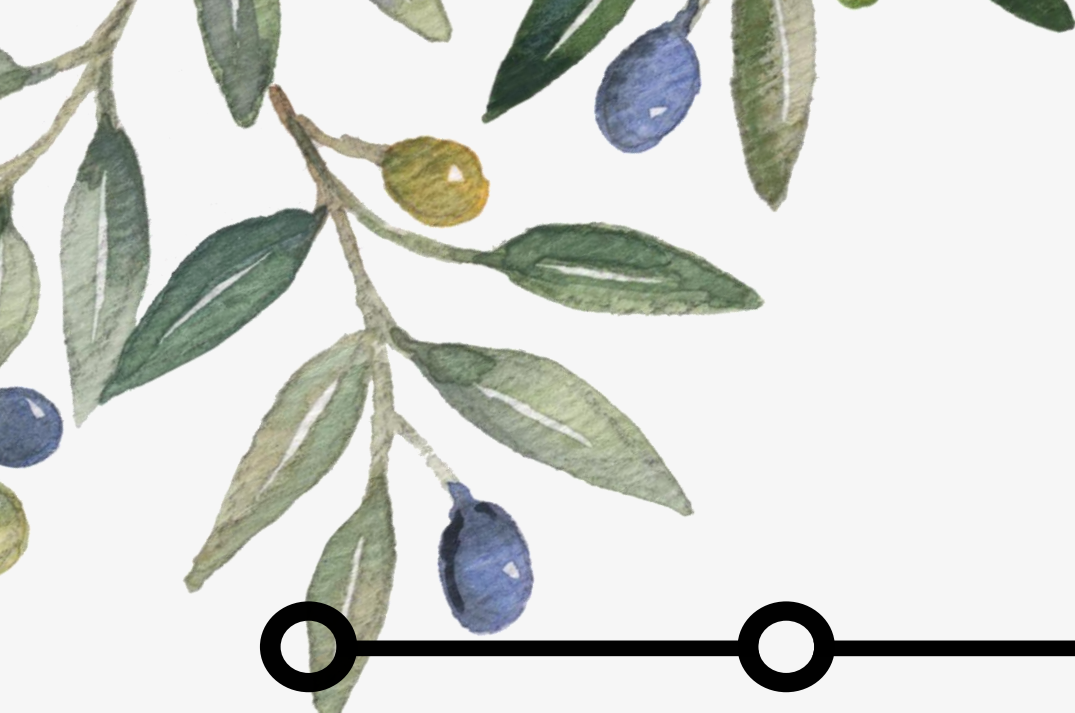
Trigger points cause referred pain, tender points do not.





Historical Development of Trigger Point Theory

- 1500s – Guillaume de Baillou (1538–1616, France) → First writings on muscle pain disorders.
- 1816 – William Balfour (1780–1846, Scotland) → “Thickenings” and “nodular tumors” linked to muscle pain.
- 1843 – Robert Remak Froriep (1804–1861, Germany) → Coins “muskelshwiele” (“muscle callouses”).
- 1904 – Sir William Richard Gowers (1845–1915, Britain) → Introduces “fibrositis.”
- 1919 – Georg Schade (Germany) → Describes nodules as “myogeloses.”
- 1938 – John Hunter Kellgren (1911–2002, Britain) → Maps referred pain with saline injections. [hypertonic](#)
- Mid-1900s – Michael Gutstein (Germany) & Michael James Kelly (Australia) → Research muscle pain syndromes.
- 1950s – Janet Graeme Travell (1901–1997, USA) & Morton Rinzler (USA) → Coin “myofascial trigger point.”
- 1983 – Travell & David G. Simons (1922–2010, USA) → Publish Myofascial Pain and Dysfunction: The Trigger Point Manual.



trigger points defined by an MD but interesting overlap osteopathy

Timeline



Janet G. Travell, M.D. (1901–1997)



- Born Dec 17, 1901, in New York City; daughter of physician John Willard Travell and Janet Davidson Travell.
- Wellesley College (1922); earned her M.D. at Cornell University Medical College (1926)
- Completed residency at New York Hospital; served as ambulance surgeon with the NYC police; research fellowship at Bellevue Hospital.

Dr. Janet Travell & John F. Kennedy

- In 1955, Senator John F. Kennedy consulted Dr. Janet Travell for severe back spasms radiating into his leg.
- She treated his lumbar myofascial trigger points with procaine injections, providing significant relief.
- Identified a leg length discrepancy and designed heel lifts for his shoes to reduce strain.
- Recommended the now-famous rocking chair to ease his back pain.
- Her care enabled Kennedy to continue his career... leading to the presidency.
- In 1961, he appointed her Personal Physician to the President, the first woman and one of the few civilians to hold the role.

Scinta, A. L. (n.d.). Dr. Janet G. Travell. Pelham Town Historian. Retrieved Sep 2025 from Pelham Town Historian website



After the White House...



- Served at George Washington University School of Medicine
- Authored over 100 scientific articles, her autobiography *Office Hours: Day and Night* (1968), and the seminal two-volume *Myofascial Pain and Dysfunction: The Trigger Point Manual* (1983).
- A trailblazing researcher, educator, and clinician...often called the “mother of trigger point medicine.” Delivered enduring contributions to pain management.
- Passed away on August 1, 1997, at her home in Northampton, Massachusetts, aged 95.



Trigger Point Definition

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Tender Points?

Trigger points cause referred pain, tender points do not.



Common Findings of Trigger Points

- Palpable taut band with cross-fiber flat or pincer palpation
- Hypersensitive spot within the taut band
- Local Twitch response when adequately stimulated
- May produce motor and autonomic phenomena
- May prevent full lengthening of the muscles
- May cause inhibition weakness of the muscles

Active Vs. Latent

ACTIVE TRPS

- Produces a patients recognized pain
- Spontaneous local or referred pain
- Reproduce any symptoms not just pain experienced
- Recognizes the symptom as familiar
- Symptom may be absent at the moment of exam, but will appear during manual palpation

LATENT TRPS

- Local or referred unrecognized pain
- Painful only when palpated
- Do not reproduce symptoms experienced by the patient
- Patient does not recognize symptoms caused by cross-fiber flat or pincer palpation

Objectives:

Identify common clinical signs and symptoms associated with trigger points.

Differentiate between pain patterns associated with specific trigger points vs radiculopathy vs counterstrain.

Compare and contrast osteopathic treatment options for trigger points and their indications and contraindications.

Explain the rationale, technique, indications, and contraindications of the Spray-and-Stretch method for treating myofascial trigger points.

Differentiating Pain Patterns

Key Takeaway:

Trigger points = non-dermatomal referral + taut band

Radiculopathy = dermatomal pattern + neurologic signs

Counterstrain = local tenderness

investigate key areas and do neuro exa

TRPT important to recognize - pts have been mis-dx with radiculopathy when just having TRPT

Myofascial Trigger Point

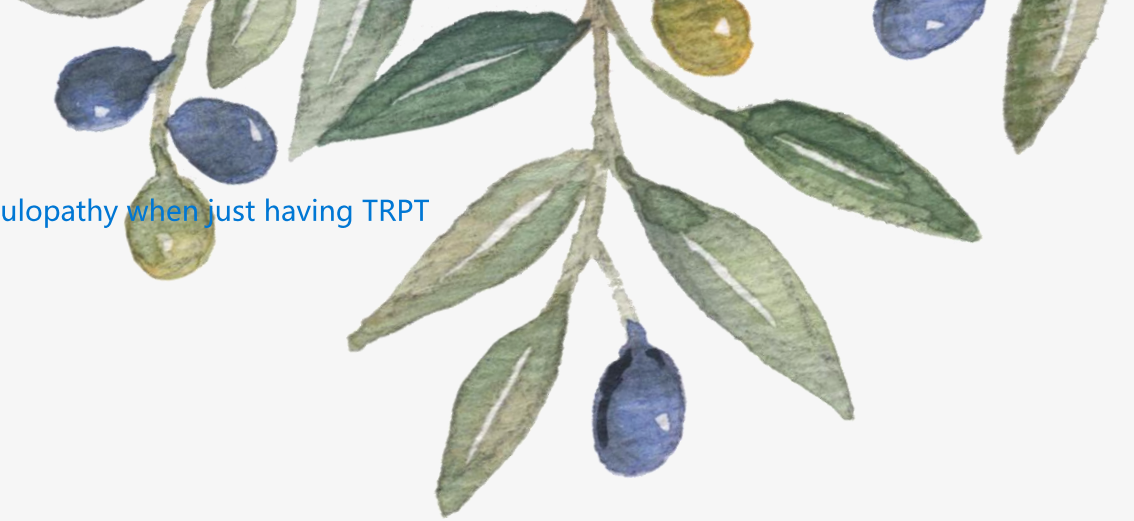
- Deep, dull, aching pain referred in a predictable but non-dermatomal pattern that reproduces the patient's familiar pain.
- Palpable taut band with a tender nodule; jump sign or local twitch when pressed. or needled
- Pain increases with stretch or sustained contraction; usually no neurologic deficits. check DTR to make sure no neurologic component

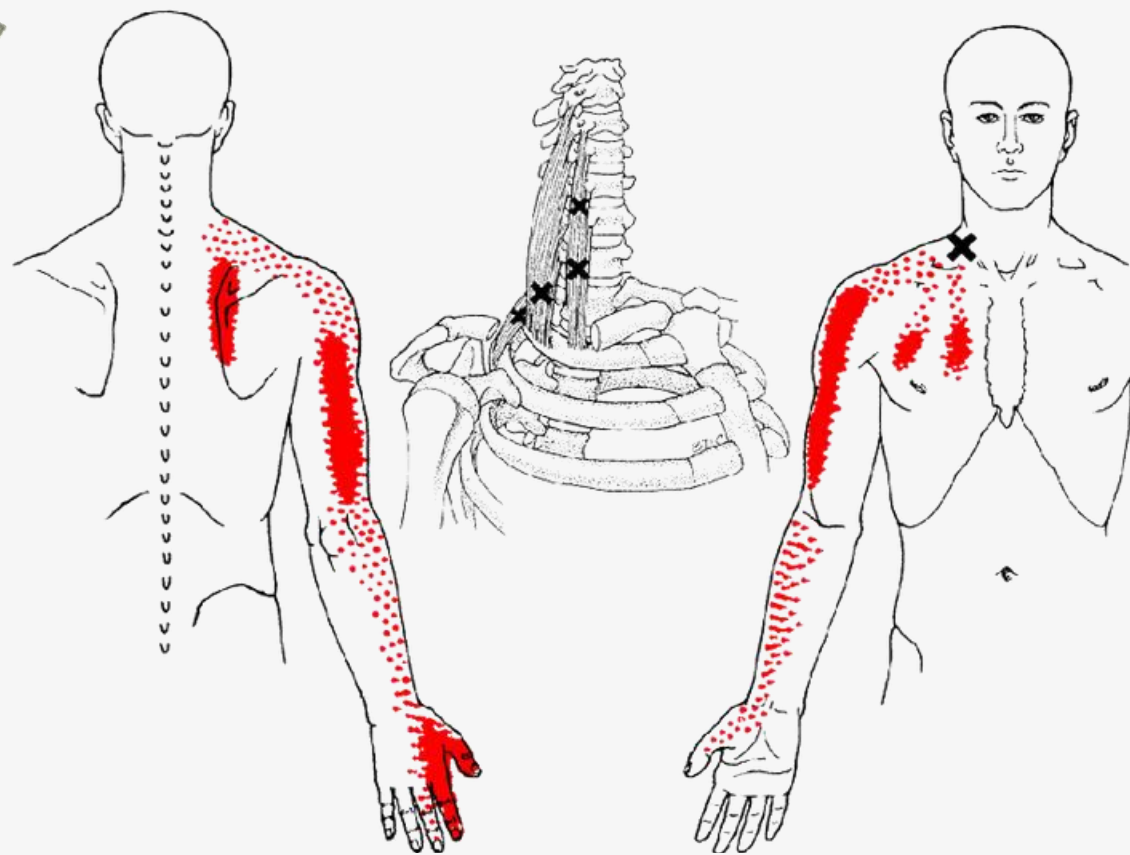
Radiculopathy

- Sharp, burning, or electric pain following a dermatomal distribution of the affected nerve root.
- Often accompanied by sensory loss, weakness, or reflex changes.
- Pain worsens with spinal movements that compress or tension the nerve.

Counterstrain Tender Point can tx cs similar to trpt - makes it more confusing

- Small, discrete area of local tenderness without referred pain.
- Pain decreases when the muscle is placed in a position of comfort.
- No neurologic findings.



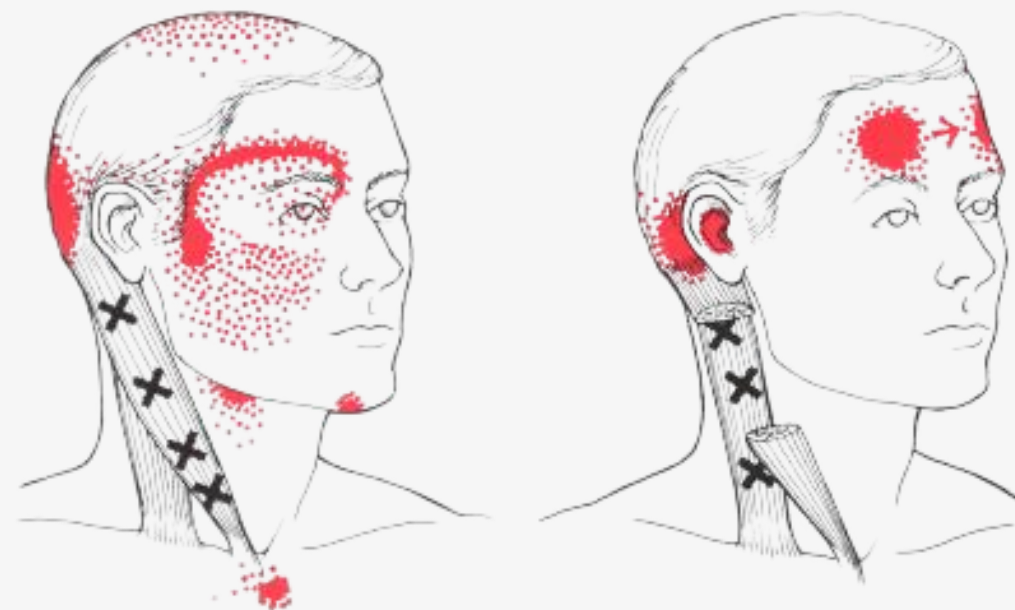


few of many

Scalenes

lateral neck into the anterolateral chest, scapular border, and down the radial arm to the thumb and index finger

check that is not disc herniation but could also be trpt



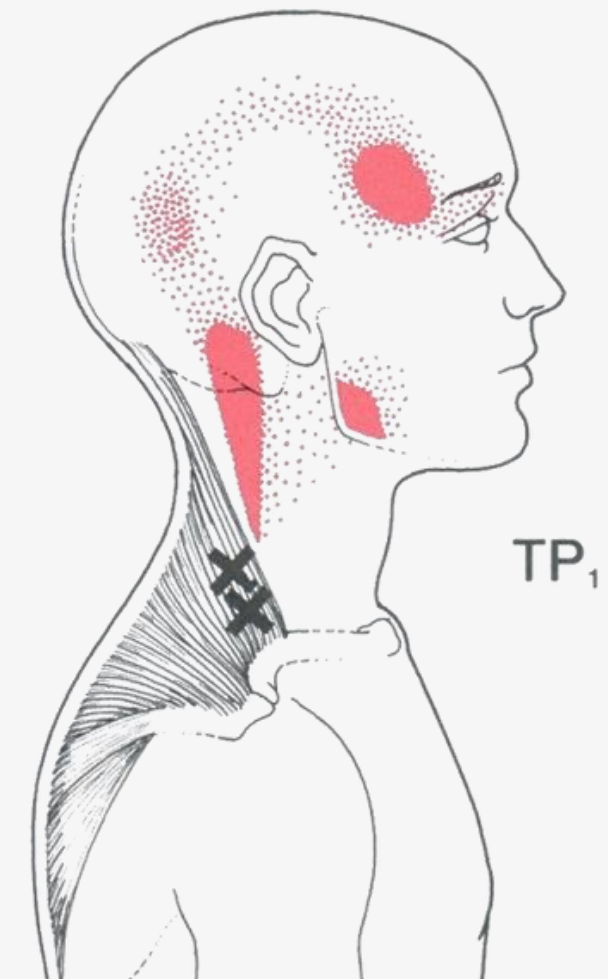
Sternocleidomastoid

unilateral around eye check SCM

referred pain to forehead, ear, and around the eye; can mimic tension or migraine headaches.

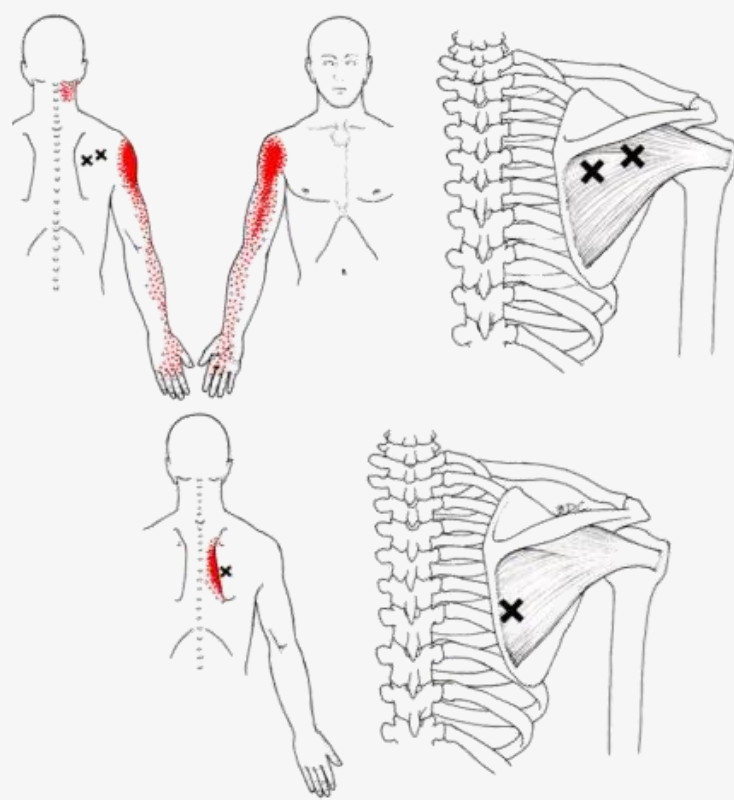
Upper Trapezius

classic source of neck/shoulder pain and tension headaches.



Common Presentations

Donnelly, J. M Myofascial Pain and Dysfunction

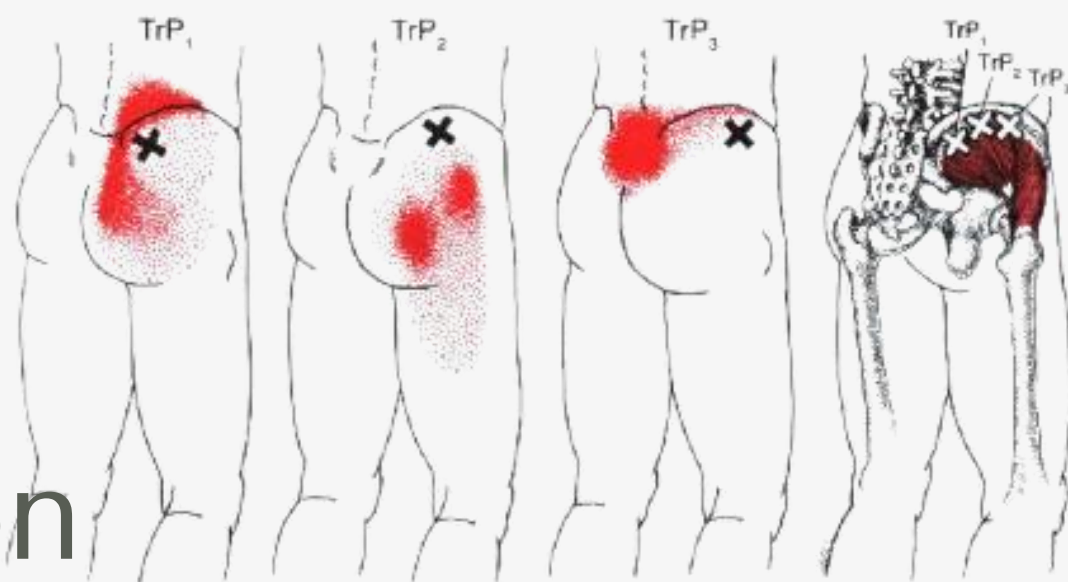
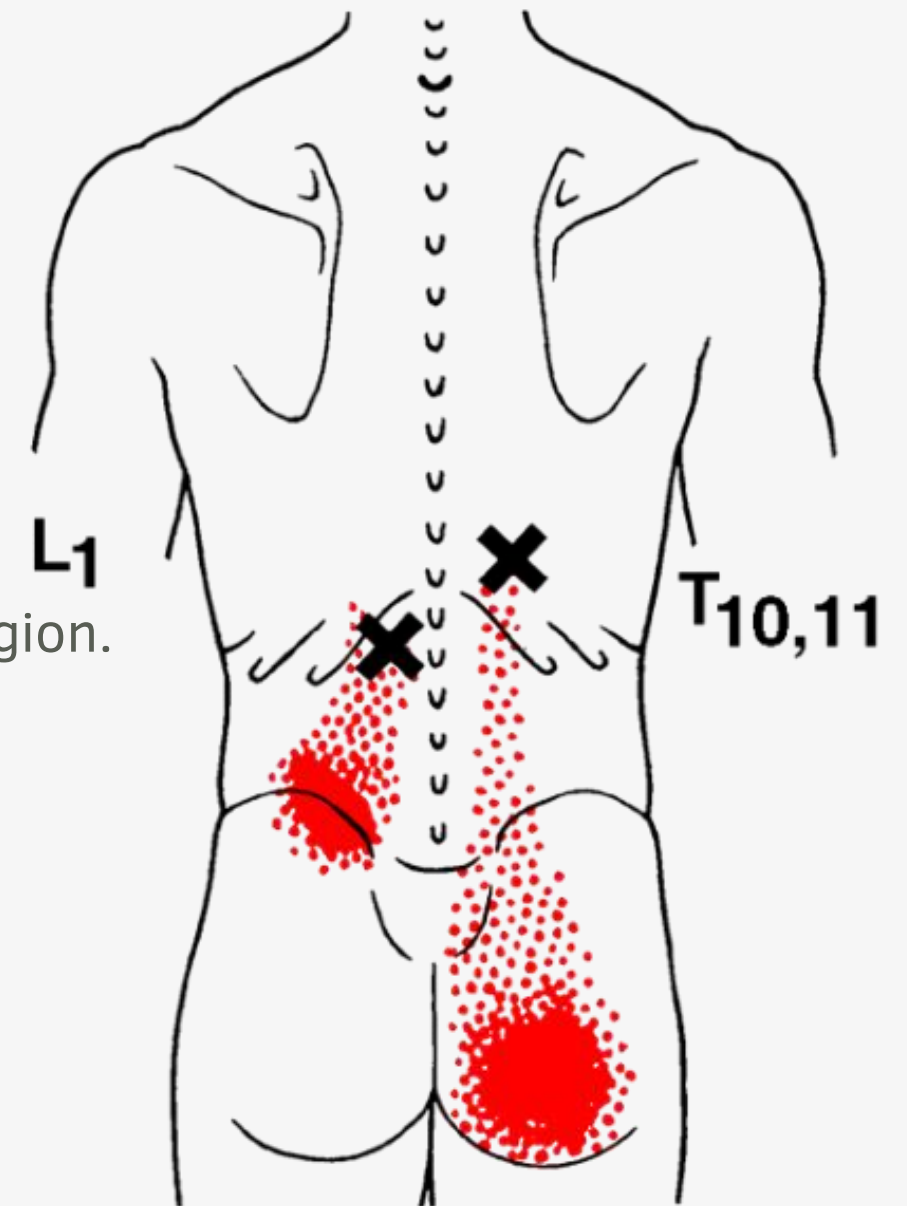


Infraspinatous

deep anterior shoulder and arm pain; often mimics rotator cuff pathology.

Quadratus Lumborum

low-back pain with referral to hip or greater trochanter region.



Gluteus medius & minimus

lateral hip and leg pain often mistaken for radiculopathy

Common Presentations

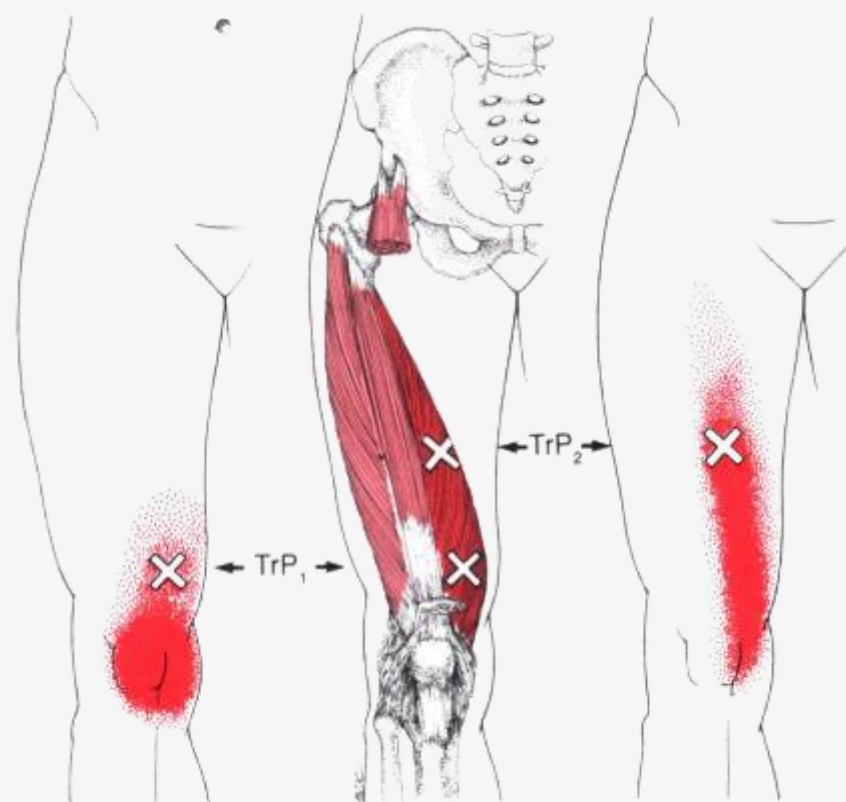
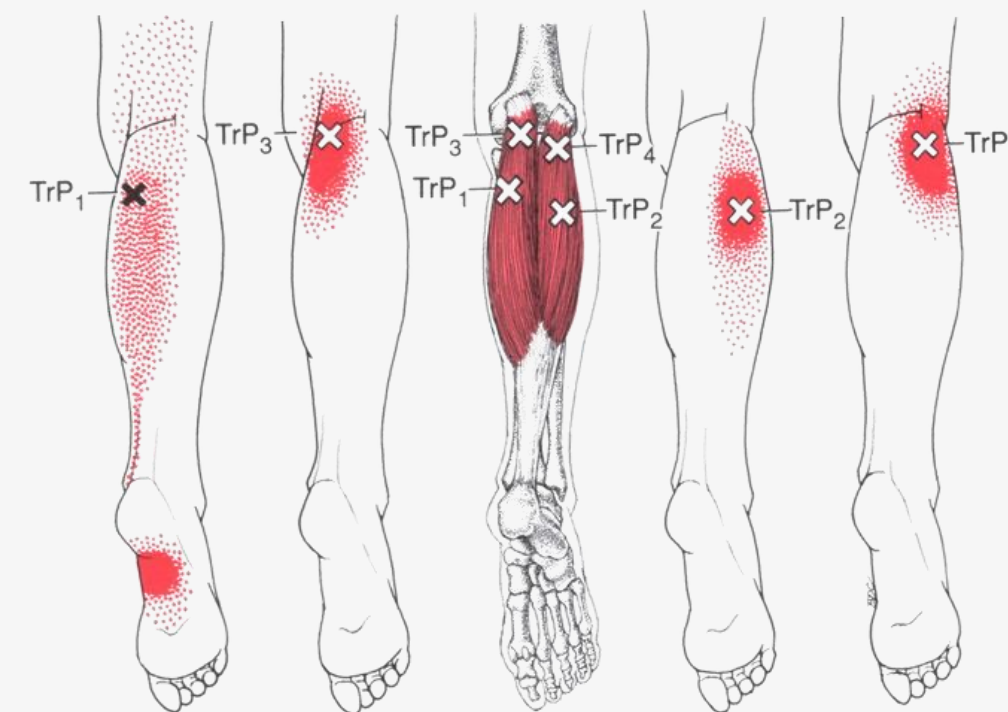
Donnelly, J. M Myofascial Pain and Dysfunction



Common Presentations

Gastrocnemius

calf pain, heel pain, night cramps.

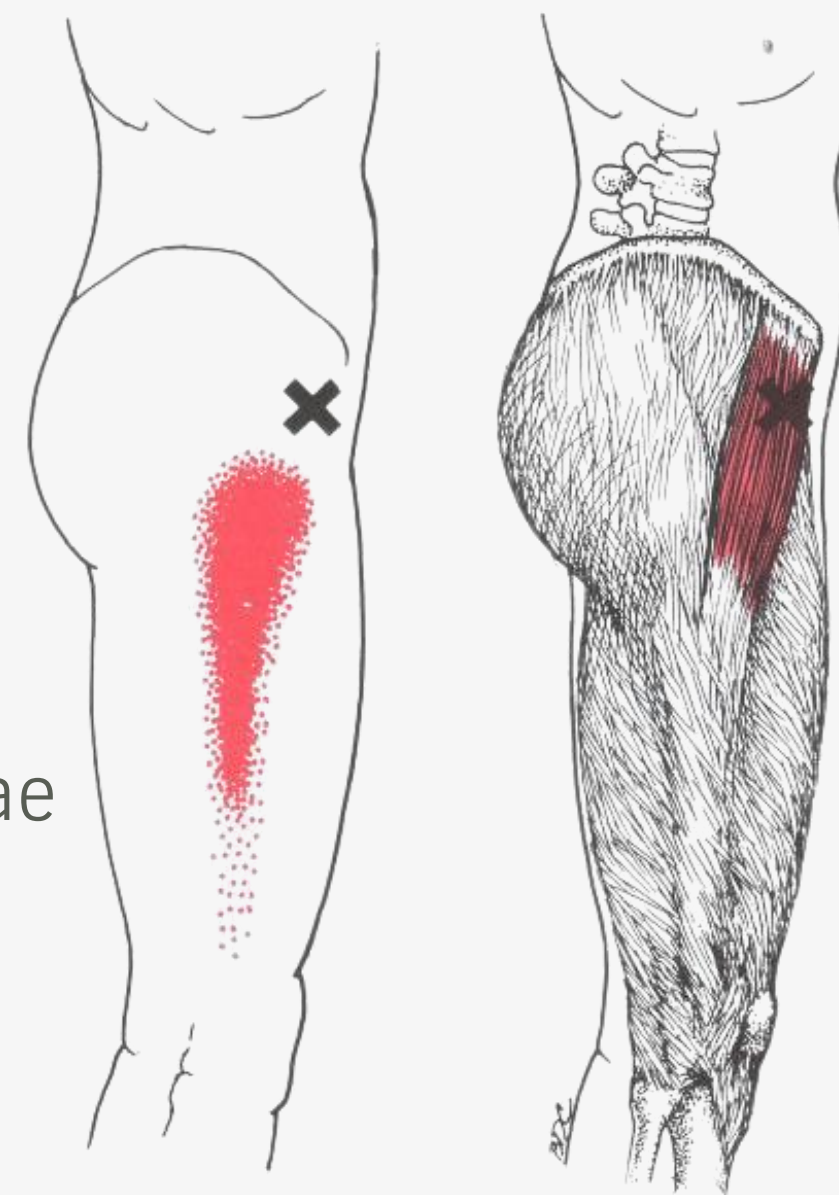


Vastus Medius

anterior knee pain, patellofemoral mimic.

Tensor Fasciae Latae

lateral thigh pain.



Osteopathic Treatments for Myofascial Trigger Points



TRIGGER POINT PRESSURE RELEASE/ISCHEMIC (DIRECT) COMPRESSION

similar to inhibition, ischemia disproven

- Mechanism: Sustained, tolerable pressure on the nodule to lengthen sarcomeres and reset muscle spindle activity.
- Indications: Active or latent trigger points causing focal or referred pain; limited range of motion.
- Contraindications: Acute fracture, local infection, open wound, severe coagulopathy, uncontrolled bleeding disorder.

can cause some bruising

MUSCLE ENERGY (POST-ISOMETRIC RELAXATION OR RECIPROCAL INHIBITION)

- Mechanism: Gentle isometric contraction followed by stretch to lengthen hypertonic fibers.
- Indications: Trigger points with associated muscle tightness and restricted motion.
- Contraindications: Acute muscle tear, severe unrelieved pain, inability to follow commands.

STRAIN–COUNTERSTRAIN

- Mechanism: Position of maximal comfort shortens the muscle, decreasing gamma gain and muscle spindle firing.
- Indications: Tender or trigger points producing protective spasm; patients intolerant of aggressive stretch or pressure.
- Contraindications: Inability to tolerate positioning (fracture, vascular compromise).

can use concept to position area to max comfort
better for a pt that can relax but can't follow directions



Osteopathic Treatments for Myofascial Trigger Points

these 2 more specific to TRPTs

MYOFASCIAL RELEASE (DIRECT OR INDIRECT)

- Mechanism: Engages fascial tension to restore tissue mobility and circulation.
- Indications: Global fascial restrictions associated with trigger points; chronic postural strain.
- Contraindications: Acute cellulitis, open wounds, deep venous thrombosis, malignancy in treatment region.

DRY NEEDLING OR TRIGGER POINT INJECTION

- Mechanism: Mechanical disruption of the taut band and chemical modulation (with or without local anesthetic).
- Indications: Refractory active trigger points unresponsive to manual therapy.
- Contraindications: Anticoagulation or bleeding disorder, local infection, patient refusal, allergy to anesthetic (if used).

acupuncture needles

SPRAY-AND-STRETCH

- Mechanism: Rapid cutaneous cooling with a vapocoolant (e.g., ethyl chloride) followed immediately by gentle passive stretch.
- Indications: Active myofascial trigger points with acute muscle spasm or restricted range of motion.
- Contraindications: Hypersensitivity or allergy to vapocoolant ingredients, cold-related disorders (Raynaud's, cold urticaria, cryoglobulinemia), open wounds or skin infection at the treatment site, impaired peripheral circulation, or inability to protect skin from excessive cold.

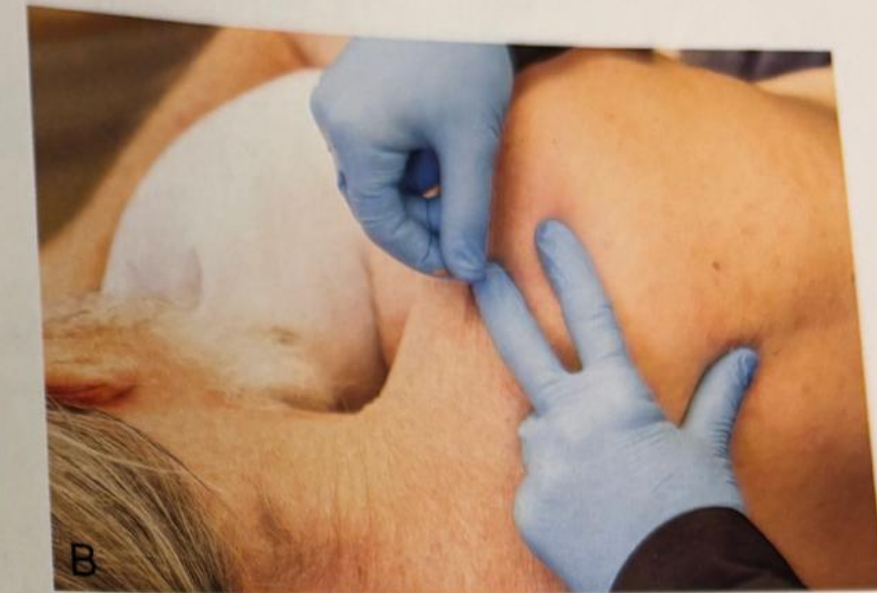
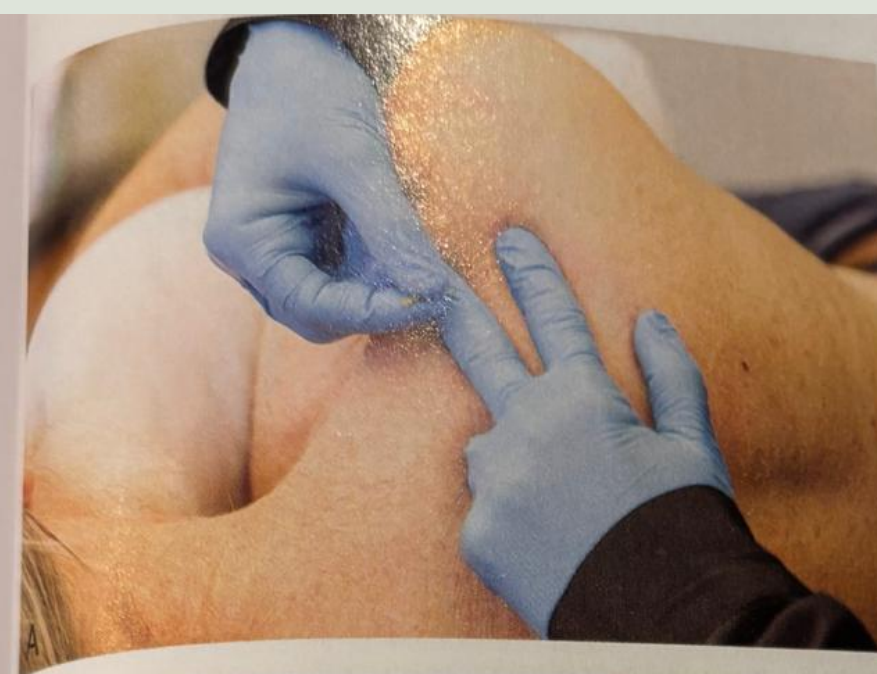
Wet Needling Vs. Dry Needling

Is there a difference?

- Dry needling (DN): solid filiform needle (Acupuncture needle) into the trigger point; no injectate.
 - Short-term pain relief vs sham; generally similar to other interventions. [PMC](#)
- Wet needling / Trigger point injection (TPI): needle plus injectate (e.g., lidocaine).
 - Some reviews suggest TPI may outperform DN in the short term, but findings are mixed.

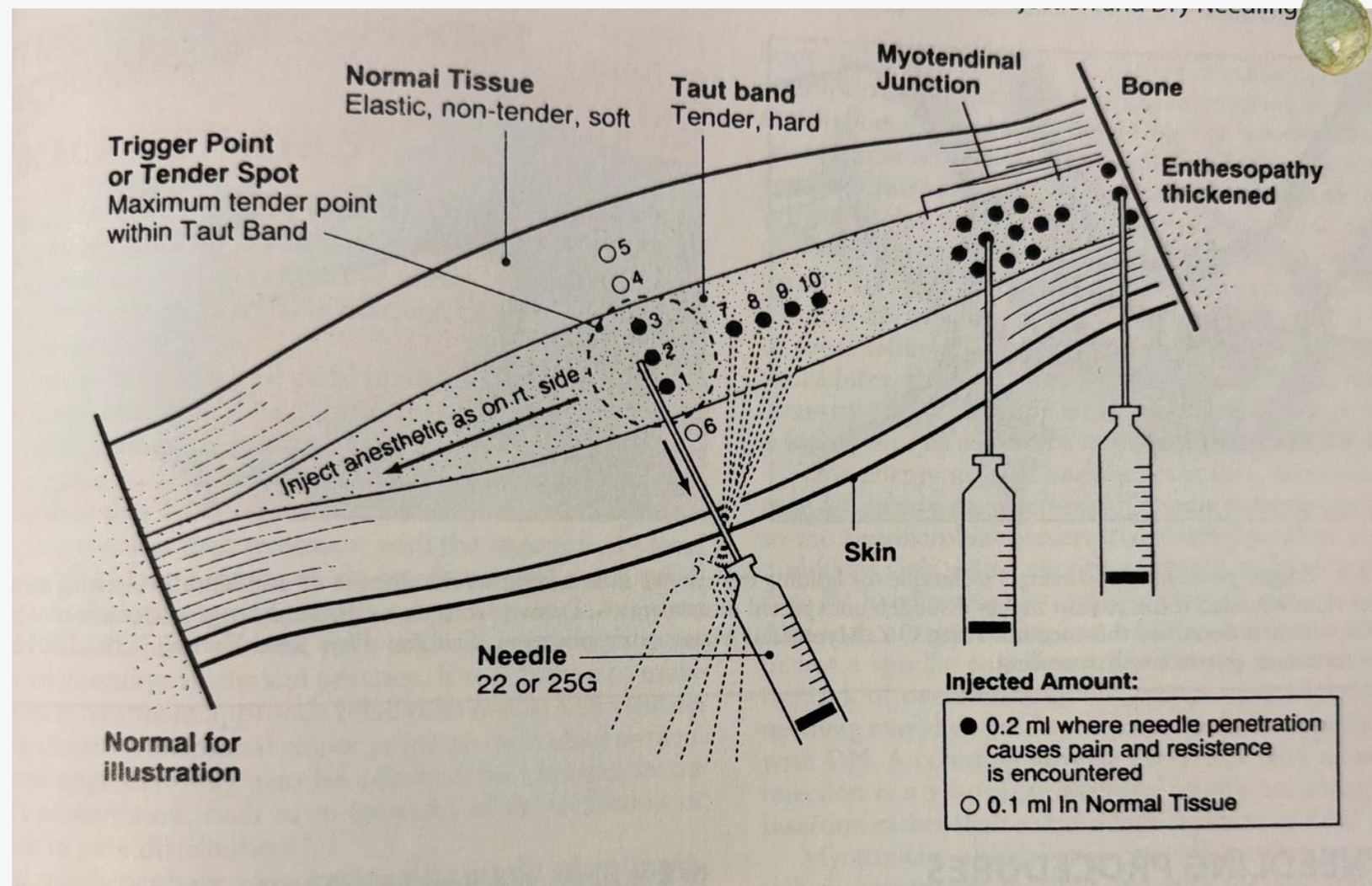
clinically M-T finds - wet needling helpful

Donnelly, J. M Myofascial Pain and
Dysfunction



supraspinatus muscle and tendon with the patient lying on the left side. A and B, the index finger of the palpating hand is on the attachment of the supraspinatus

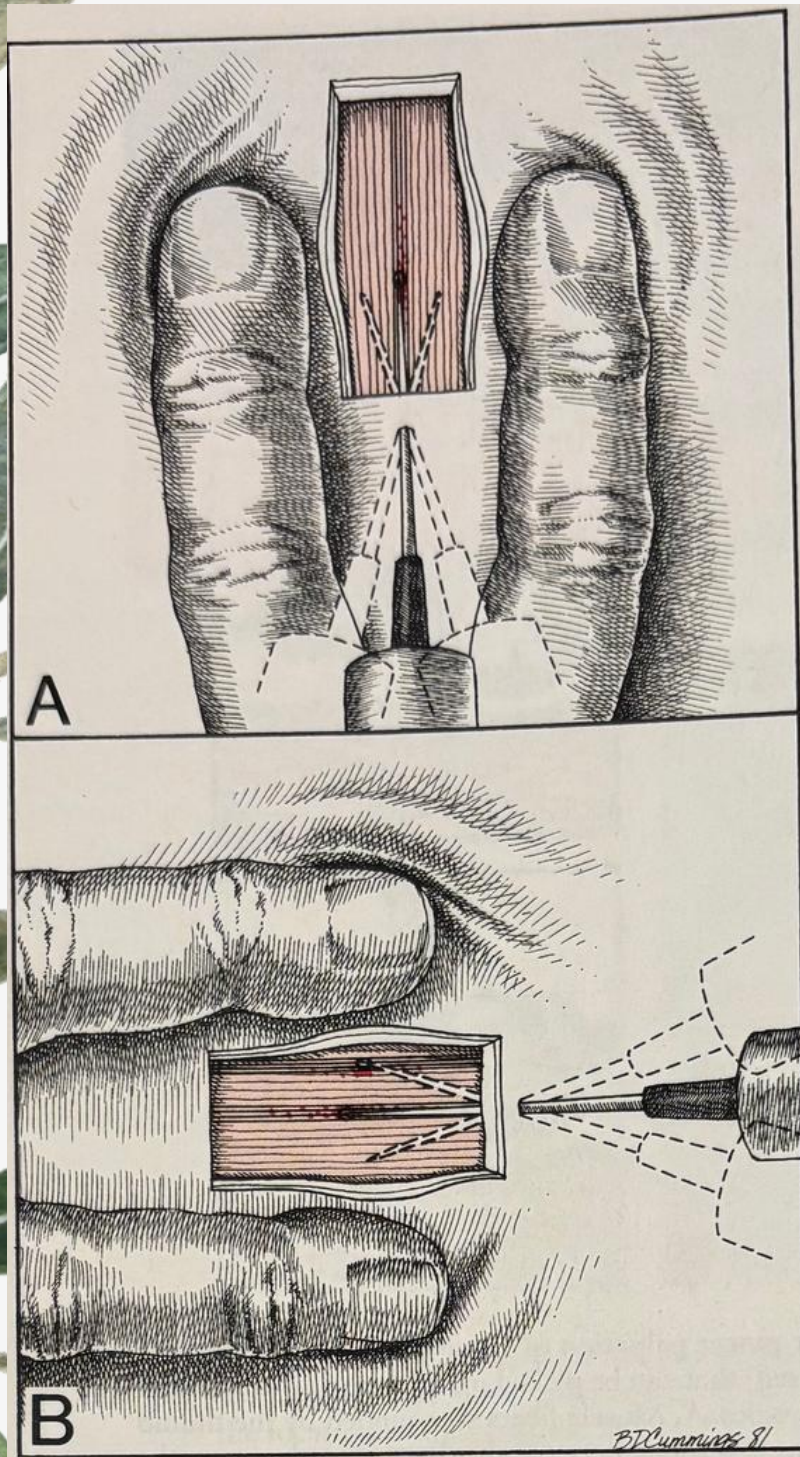
What gets injected?



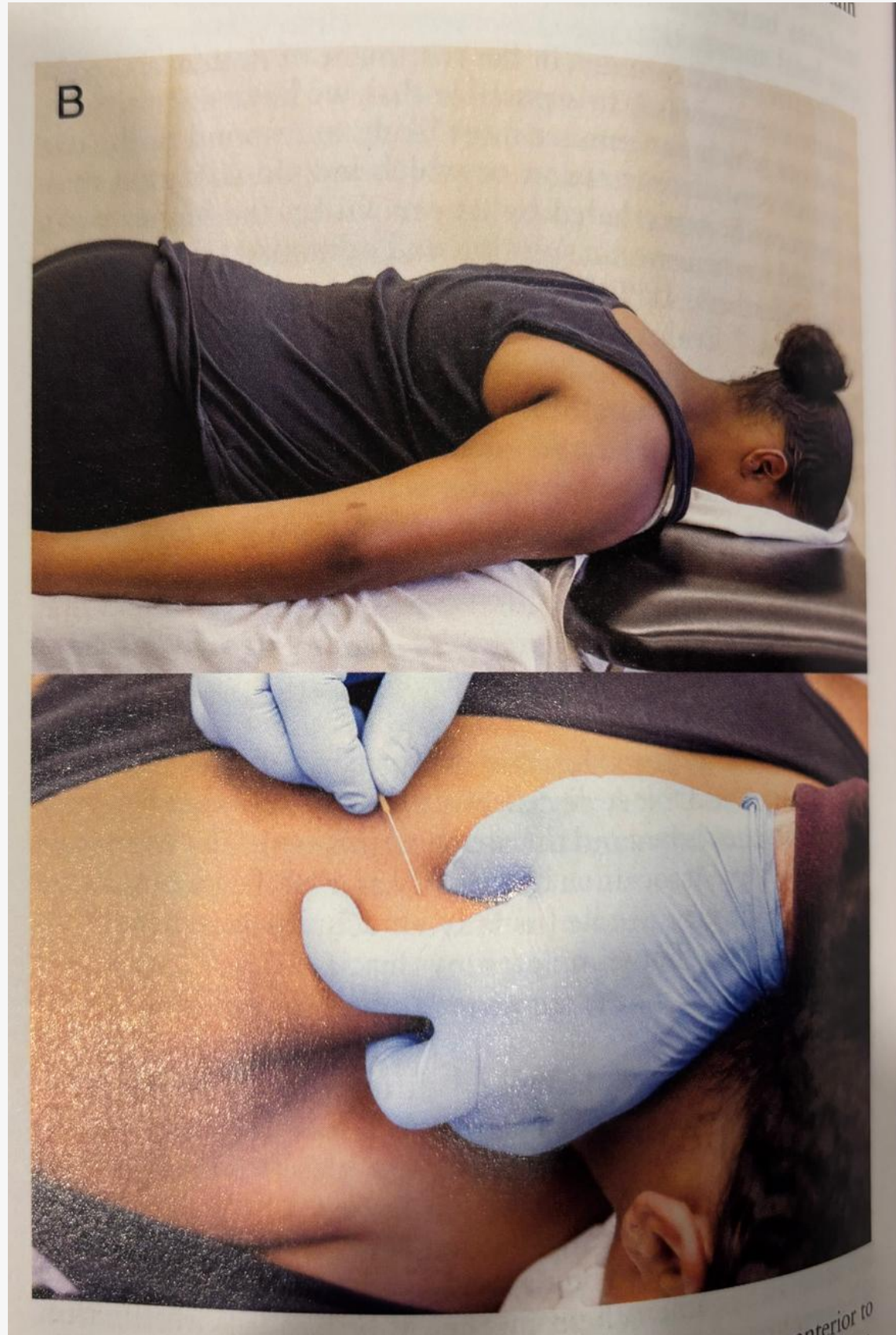
Donnelly, J. M Myofascial Pain and Dysfunction

- Local anesthetic: lidocaine (0.5–1%), mepivacaine, bupivacaine.
- Normal saline or sterile water.
- Other/less common: corticosteroid (e.g., triamcinolone), botulinum toxin A, sodium bicarbonate, ozone, hyaluronidase.
- [PRP or prolo](#)
- Key point: No single pharmacologic agent has proven superior to others or to placebo.

Which works better: steroid, lidocaine, or dry needling?



- Lidocaine TPI vs DN:
 - Similar clinical improvement in several small RCTs; less post-needling soreness with lidocaine. ([Hong, Eroğlu](#))
- Lidocaine vs saline:
 - Both reduce pain at 2–4 weeks; lidocaine may have a faster 10-minute analgesic effect. ([Tantanatip](#))
- Steroid + anesthetic vs anesthetic alone:
 - For trigger points, high-quality evidence does not show clear added benefit of adding steroid; guidelines note no agent is superior overall. (Misirlioglu, Saleh)



Post Needling Procedures

Immediate Care

- Apply firm digital pressure over each needling site for 30–60 seconds to limit bleeding and local soreness.
- If bleeding is minimal, apply a small bandage only if needed.

Active Movement

- Immediately after treatment, the patient actively moves the treated muscle through its full range of motion three times,
 - reaching its fully shortened position
 - then its fully lengthened position each cycle.
- Purpose: distributes local anesthetic (if used), restores normal sarcomere length, and reduces post-needling stiffness.



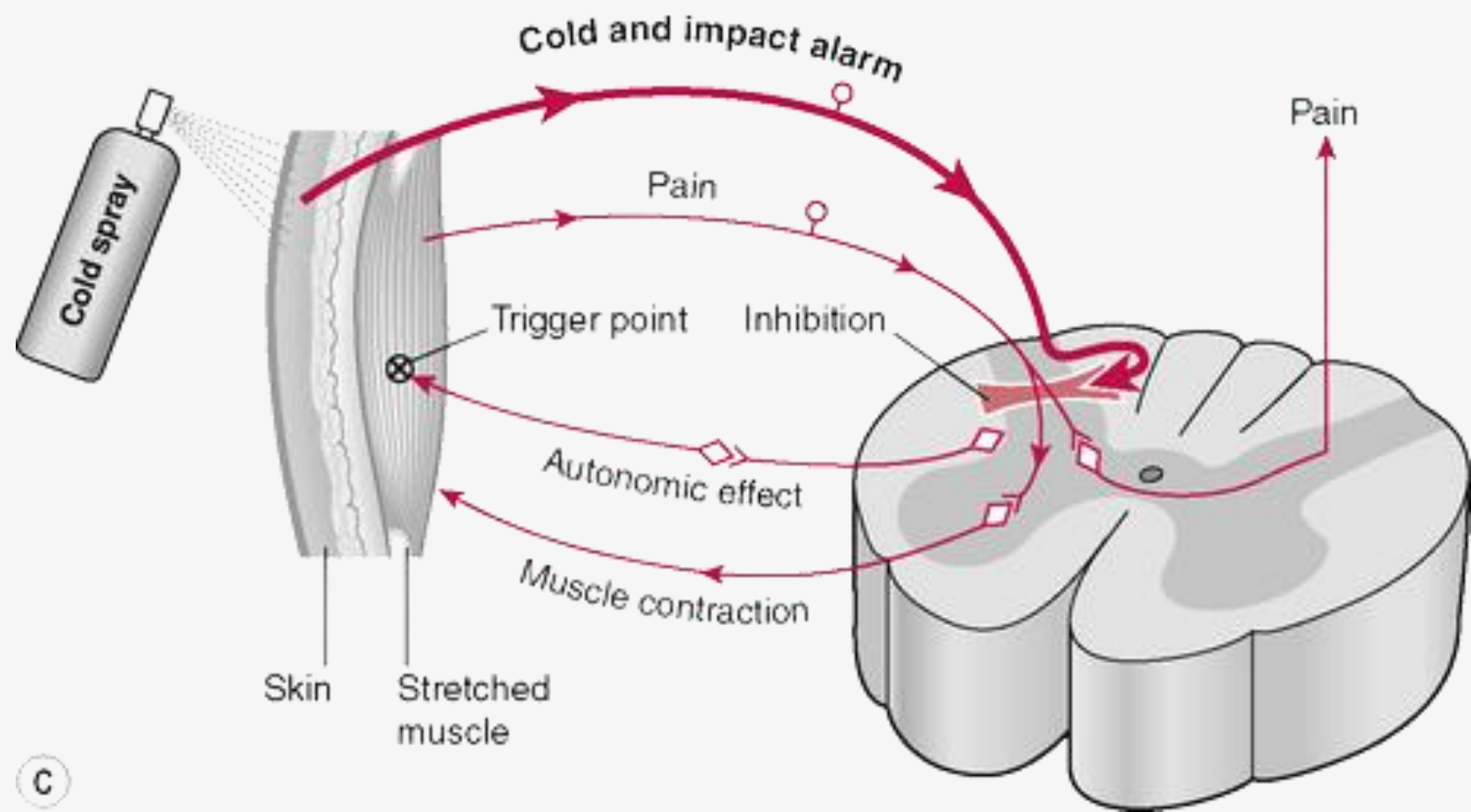
Spray And Stretch

- Developed by Simons et al 1983 as a non-invasive method to inactivate myofascial trigger points.
- Combines cutaneous cooling with gentle passive stretch to reduce pain, interrupt the pain-spasm-pain cycle, and lengthen shortened muscle fibers.

Can be used for single muscles but also for complex multimuscular regions such as the shoulder the interacting trigger points can be worked with together with spray and stretch increasing the function of the whole regional group.



Spray and Stretch



Cryotherapy counteracts the change in pain threshold. Decreasing nerve conduction as well as nociceptive threshold, requiring more noxious input for perception of pain.

Cryotherapy can decrease inflammatory changes reducing histamine, mast cells, substance P, calcitonin, bradykinin, and others which can potentially prevent central sensitization. Reduction in nociception allows for stretch without input from trigger point.

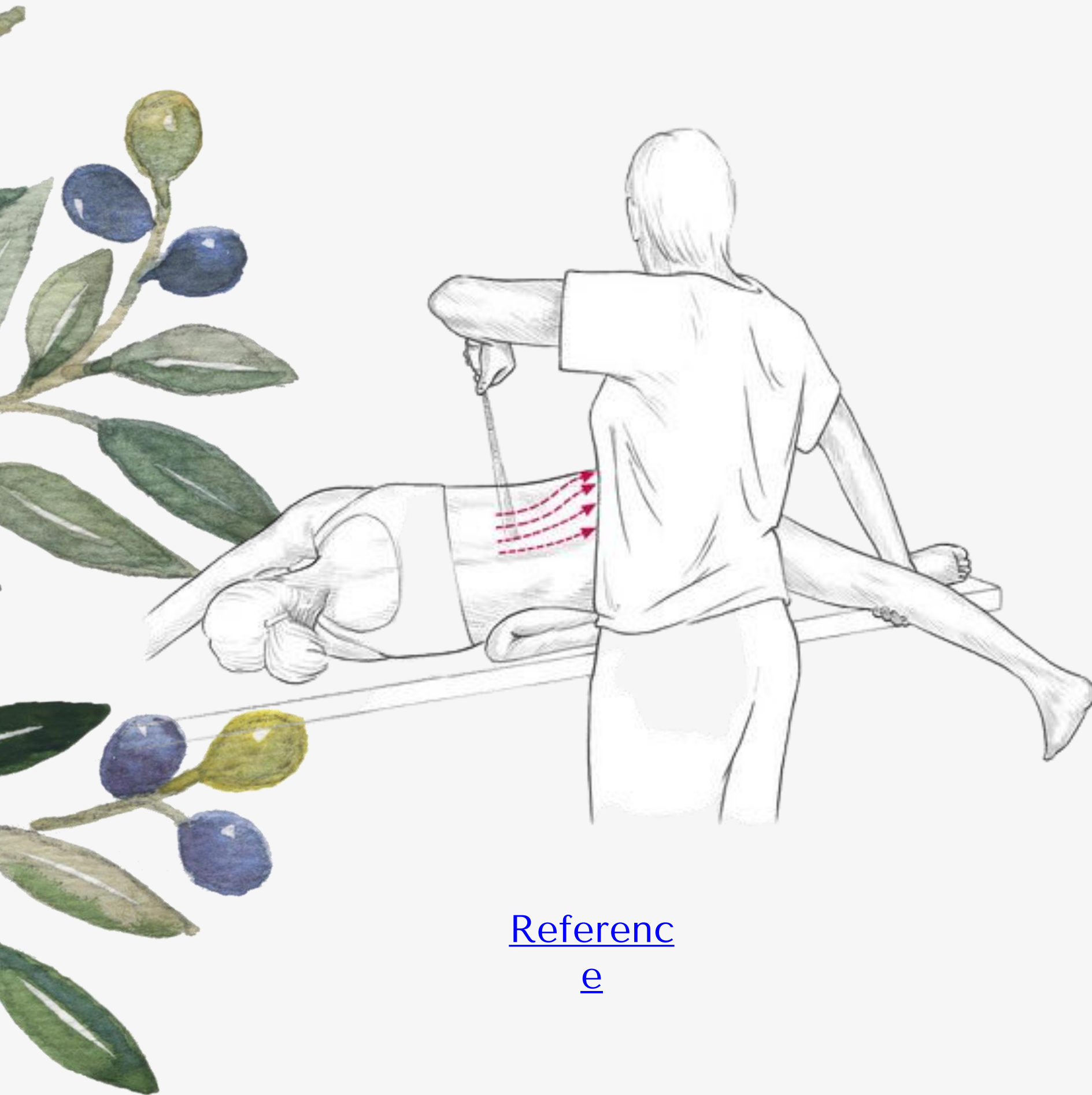
[Reference](#)
[e](#)

Spray and Stretch

Vapocoolant spray applied directly to skin, stream from 12-18 cm away from the skin at a 30 degree angle.

Physician performs spray of entire muscle length and pain referral area two or three times and then immediately place that muscle on stretch.

[Reference](#)
[e](#)



Spray And Stretch

Indications

- Active or latent myofascial trigger points causing:
- Local or referred myofascial pain
- Muscle tightness or spasm
- Limited range of motion
- Acute muscle strain where guarding prevents direct pressure
- Postural or repetitive-strain syndromes with shortened muscle groups
- Patients intolerant of needling or injection, or when invasive therapy is contraindicated
- Adjunct to rehabilitation: to allow pain-free stretching before strengthening or corrective exercise

Contraindications

- Hypersensitivity or allergy to vapocoolant ingredients
- Cold-related disorders:
 - Cold urticaria
 - Cryoglobulinemia
 - Raynaud's phenomenon
- Compromised skin integrity at treatment site
 - Open wounds
 - Dermatitis, infection, or recent surgery
- Impaired circulation or peripheral vascular disease where cold application may worsen ischemia
- Uncontrolled neurologic or systemic disease where sudden cold exposure may trigger adverse reactions (e.g., severe autonomic instability)



Key Take Aways



Trigger Point Essentials

- Definition: Hyperirritable spot in a taut band of skeletal muscle that produces characteristic, non-dermatomal referred pain and restricted motion.
- Common Sites: Upper trapezius, levator scapulae, SCM, infraspinatus, gluteus medius/minimus, quadratus lumborum.

Diagnosis

- Palpable taut band & tender nodule, jump sign, reproducible referred pain, possible local twitch response.
- Differentiate from radiculopathy (dermatomal + neuro deficits) and counterstrain tender points (local tenderness only).

Treatment Spectrum (OMT & Adjuncts)

- Manual: Ischemic compression/Pressure Release, Muscle Energy, Strain-Counterstrain, Myofascial Release.
- Procedural: Dry needling, Trigger Point Injection (lidocaine or saline), Spray-and-Stretch.
- Evidence shows no clear advantage of adding corticosteroid to local anesthetic.

Post-Needling Care

- Immediate firm pressure.
- Active full range of motion—shortened to lengthened—×3 cycles.
- Gentle stretch to reduce soreness.

Big Picture

- Address contributing factors: posture, ergonomics, stress, sleep, systemic issues.
- Combine patient education, self-stretching, and lifestyle modification with OMT for durable results.



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